

ARCHITECTURE DIAGRAMS & PAPER REFERENCES:

VGG-19 Architecture:

Paper: Very Deep Convolutional Networks for Large-Scale Image Recognition

Authors: K. Simonyan, A. Zisserman (2014)

Reference: <https://arxiv.org/abs/1409.1556>

Key Features:

- 19 layers deep (16 conv + 3 FC)
- 3x3 convolution filters throughout
- 2x2 max pooling with stride 2
- Fixed 224x224 input size

Input(224x224x3) → [Conv64]x2 → MaxPool → [Conv128]x2 → MaxPool
→ [Conv256]x4 → MaxPool → [Conv512]x4 → MaxPool → [Conv512]x4 →
MaxPool → FC4096 → FC4096 → FC2 → Output

ResNet-50 Architecture:

Paper: Deep Residual Learning for Image Recognition

Authors: K. He, X. Zhang, S. Ren, J. Sun (2015)

Reference: <https://arxiv.org/abs/1512.03385>

Key Features:

- Residual connections for gradient flow
- Bottleneck blocks (1x1, 3x3, 1x1 conv)
- 50 layers deep
- Pre-trained on ImageNet with 1.28M images

Input → Conv7x7 → MaxPool → [Conv Block]x3 → [Identity Block]x3 → [Conv
Block]x4 → [Identity Block]x4 → [Conv Block]x6 → [Identity Block]x6
→ [Conv Block]x3 → [Identity Block]x3 → AvgPool → FC1000 → FC2

Inception-V1 (GoogLeNet):

Paper: Going Deeper with Convolutions

Authors: C. Szegedy et al. (2014)

Reference: <https://arxiv.org/abs/1409.4842>

Key Features:

- Inception modules with parallel convolutions
- Auxiliary classifiers for training
- 22 layers deep but computationally efficient
- Winner of ILSVRC 2014

Input → Stem → Inception(3a) → Inception(3b) → MaxPool → Inception(4a) → Aux1 → Inception(4b) → Inception(4c) → Inception(4d) → Aux2 → Inception(4e) → MaxPool → Inception(5a) → Inception(5b) → AvgPool → FC2

MobileNet-V3 Large:

Paper: Searching for MobileNetV3

Authors: A. Howard et al. (2019)

Reference: <https://arxiv.org/abs/1905.02244>

Key Features:

- Network architecture search (NAS)
- Hard-swish activation function
- Squeeze-and-excitation blocks
- Optimized for mobile/edge devices

Input → Conv2d 3×3 → Bottleneck (16, 3×3) → Bottleneck (16, 3×3) → Bottleneck (24, 3×3) → Bottleneck (24, 3×3) → Bottleneck (40, 5×5) → Bottleneck (40, 5×5) → Bottleneck (40, 5×5) → Bottleneck (80, 3×3) → Bottleneck (80, 3×3) → Bottleneck (80, 3×3) → Bottleneck (80, 3×3) → Bottleneck (112, 3×3) → Bottleneck (112, 3×3) → Bottleneck (160, 5×5) → Bottleneck (160, 5×5) → Bottleneck (160, 5×5) → Conv2d 1×1 → Pool → FC2

MODEL COMPARISON ANALYSIS:

VGG-19 (from scratch):

Pros:

- Simple architecture, easy to understand
- No pretraining needed (from scratch)
- Good baseline for comparison

Cons:

- Very large number of parameters (~420M)
- Slow training and inference

- Prone to overfitting without proper regularization

Best Use Case: Educational purposes, baseline models, when no pretrained models available

ResNet-50 (transfer learning):

Pros:

- Residual connections prevent vanishing gradients
- Pre-trained on ImageNet (good feature extractor)
- Relatively fast convergence

Cons:

- Still relatively large model
- May overfit small medical datasets
- Requires careful fine-tuning strategy

Best Use Case: General-purpose medical imaging, good balance of accuracy and speed

Inception-V1 (GoogLeNet):

Pros:

- Computationally efficient (Inception modules)
- Multiple scales in single layer
- Auxiliary classifiers help with gradient flow

Cons:

- Complex architecture (harder to implement)
- Auxiliary outputs require special handling
- Less commonly used than newer architectures

Best Use Case: When computational efficiency is important, multi-scale feature learning

MobileNet-V3 (transfer learning):

Pros:

- Extremely efficient (designed for mobile)
- Hardswish activation (better than ReLU)
- Network architecture search optimized

Cons:

- Lower absolute accuracy than larger models
- May struggle with fine details in medical images
- Less pretraining data specialized for medical tasks

Best Use Case: Edge deployment, real-time applications, limited computational resources

TRAINING PERFORMANCE COMPARISON:

ResNet-50 (transfer)	Final: 96.92%	Best: 96.92%
MobileNet-V3 (transfer)	Final: 96.75%	Best: 96.75%
VGG-19 (from scratch)	Final: 92.82%	Best: 92.82%
Inception-V1 (transfer)	Final: 84.10%	Best: 88.89%

EVALUATING MODELS ON TEST SET:

VGG-19 (from scratch):

Accuracy: 0.9438
Precision: 0.9281
Recall: 0.9298
F1-Score: 0.9290

ResNet-50 (transfer):

Accuracy: 0.9574
Precision: 0.9438
Recall: 0.9491
F1-Score: 0.9464

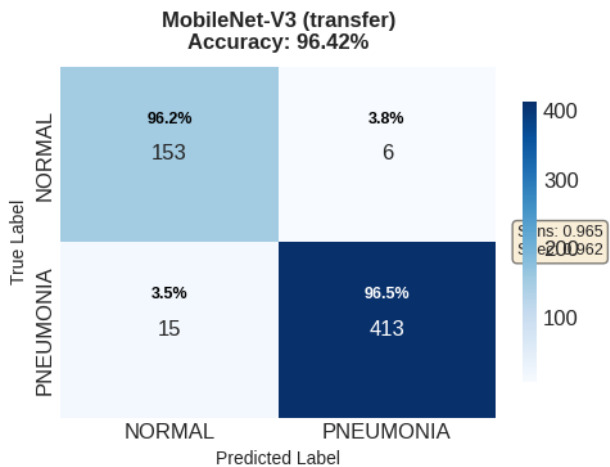
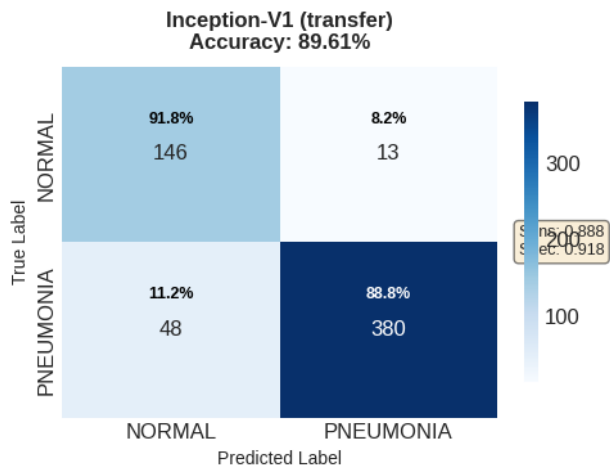
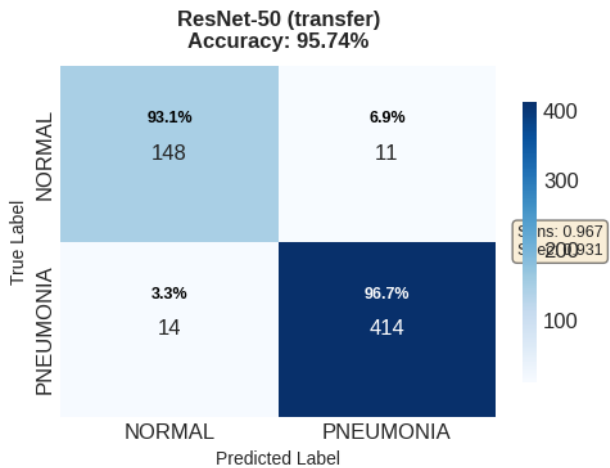
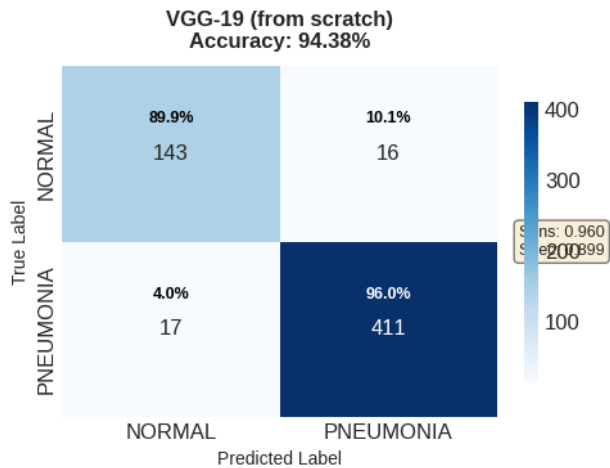
Inception-V1 (transfer):

Accuracy: 0.8961
Precision: 0.8597
Recall: 0.9030
F1-Score: 0.8764

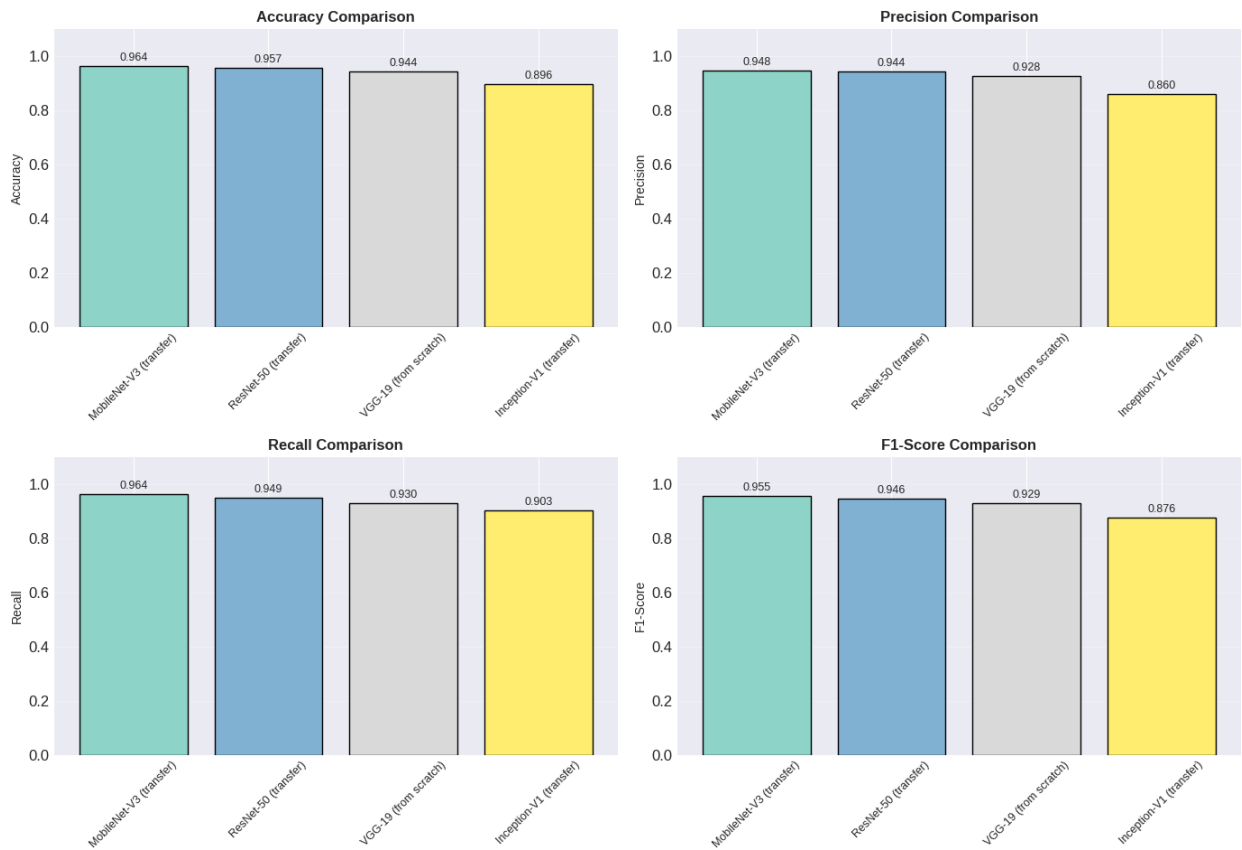
MobileNet-V3 (transfer):

Accuracy: 0.9642
Precision: 0.9482
Recall: 0.9636
F1-Score: 0.9555

Model Confusion Matrices Comparison



Model Performance Comparison Across Metrics



MODEL PERFORMANCE COMPARISON TABLE

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	Model	Accuracy	Precision	Recall	F1-Score
	MobileNet-V3 (transfer)	0.964225	0.948197	0.963609	0.955493
	ResNet-50 (transfer)	0.957411	0.943849	0.949054	0.946405
	VGG-19 (from scratch)	0.943782	0.928140	0.929826	0.928978
	Inception-V1 (transfer)	0.896082	0.859749	0.903045	0.876448

Model Performance Radar Chart
(Higher is Better)

